

Lecture 8

Radiation Distributions and Damping

Supplemental Figures

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Synchrotron radiation emitted around the ring

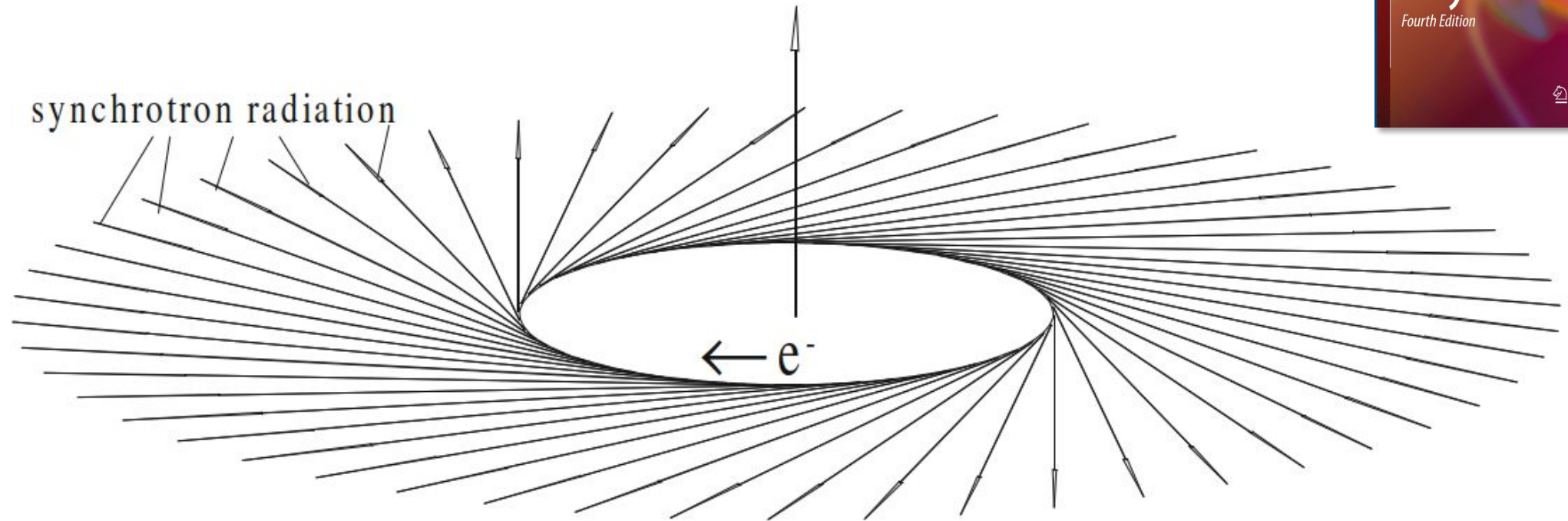
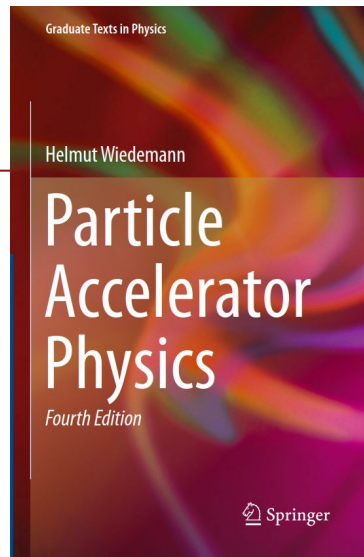


Fig. 25.6 Synchrotron radiation from a circular particle accelerator

Wavelength shifter / half-period wiggler

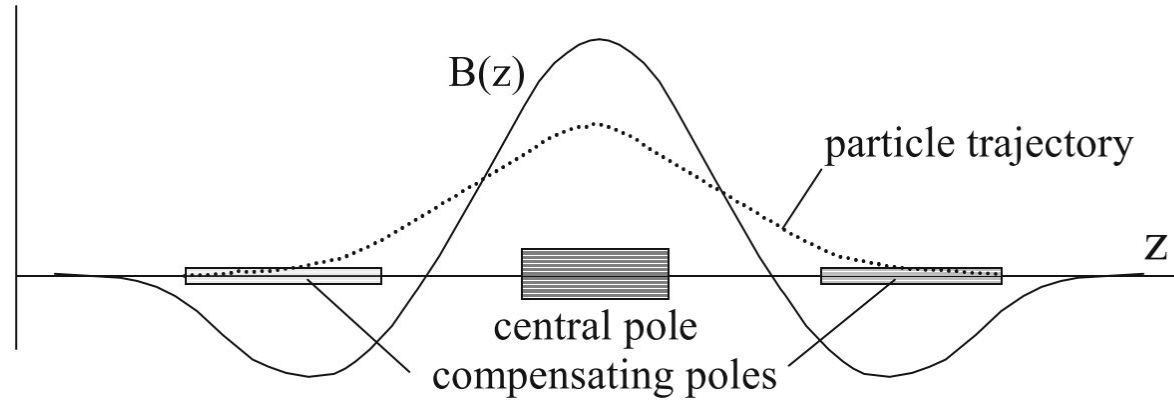
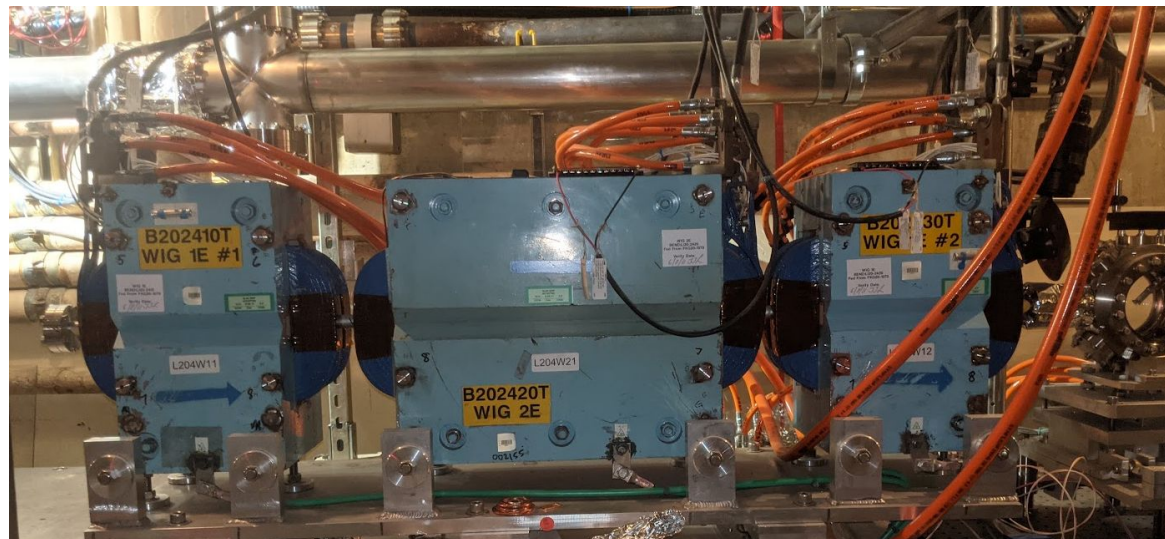
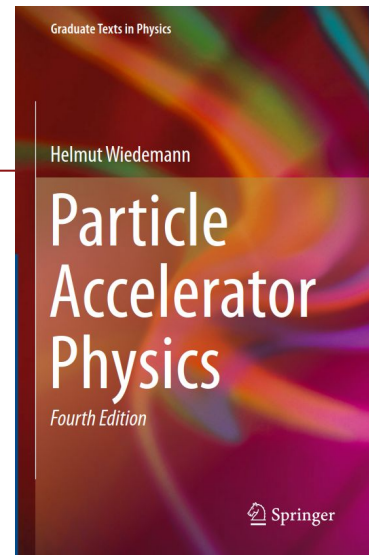


Fig. 24.2 Magnetic field distribution along the beam path for a wave length shifter



“Wiggler” at FACET-II for a synchrotron radiation diagnostic



Wiggler

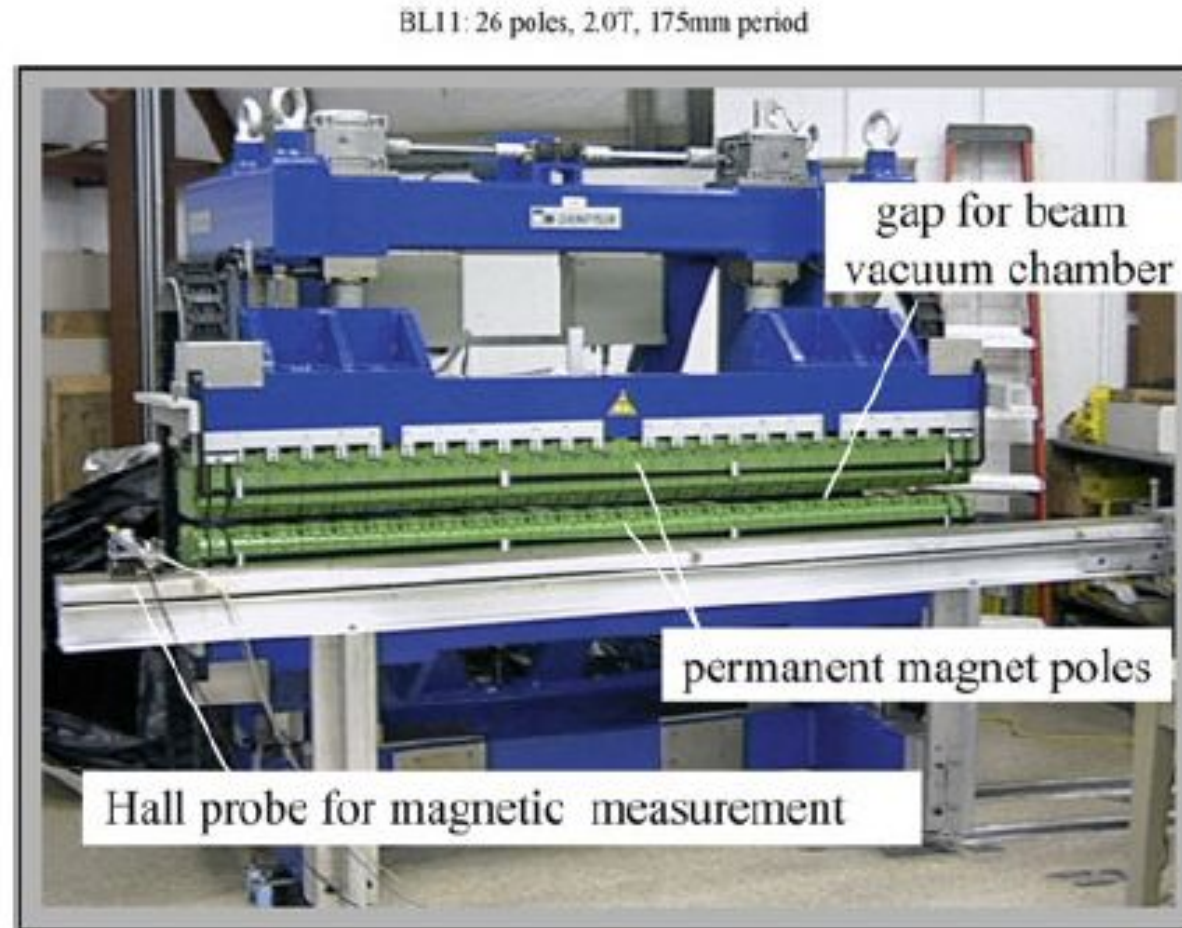
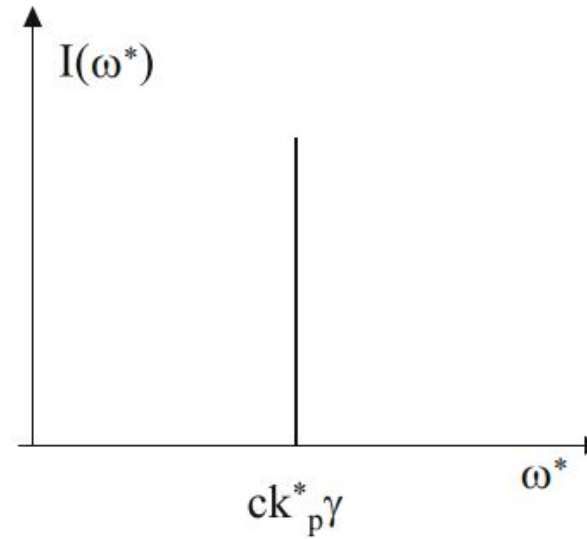
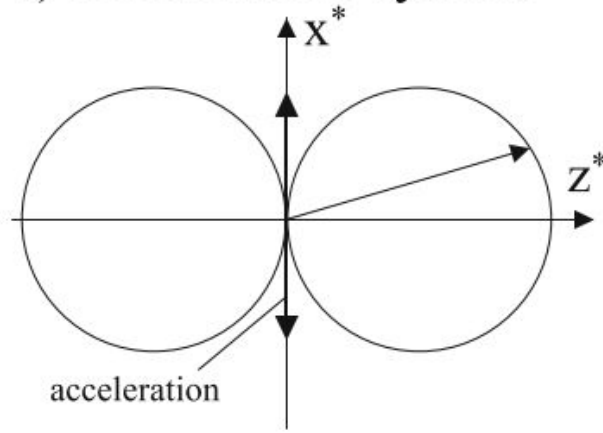


Fig. 24.4 Permanent magnet wiggler magnet with 26 poles, a 175 mm period length and a maximum field of 2.0 T



Weak undulator radiation

a) electron rest system



b) laboratory system

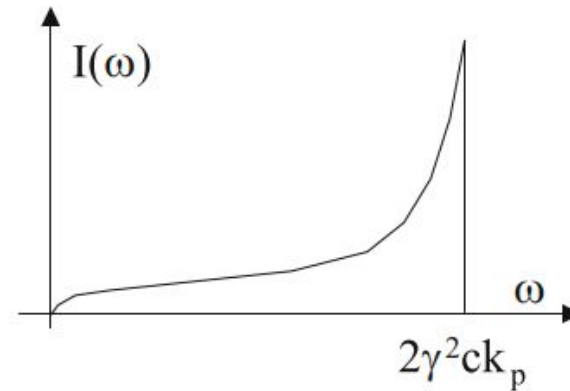
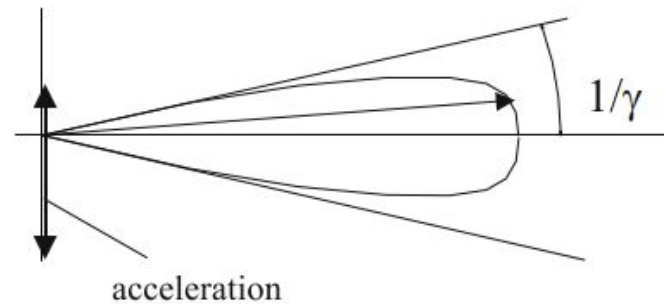
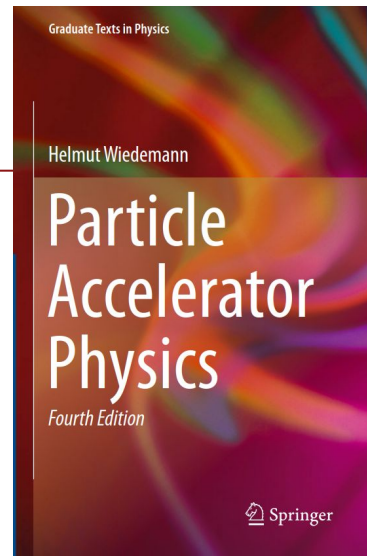


Fig. 24.6 Beam dynamics and radiation lobes in the particle rest system (a) and the laboratory system (b) for a weak undulator ($K \ll 1$)



Strong(er) undulator radiation

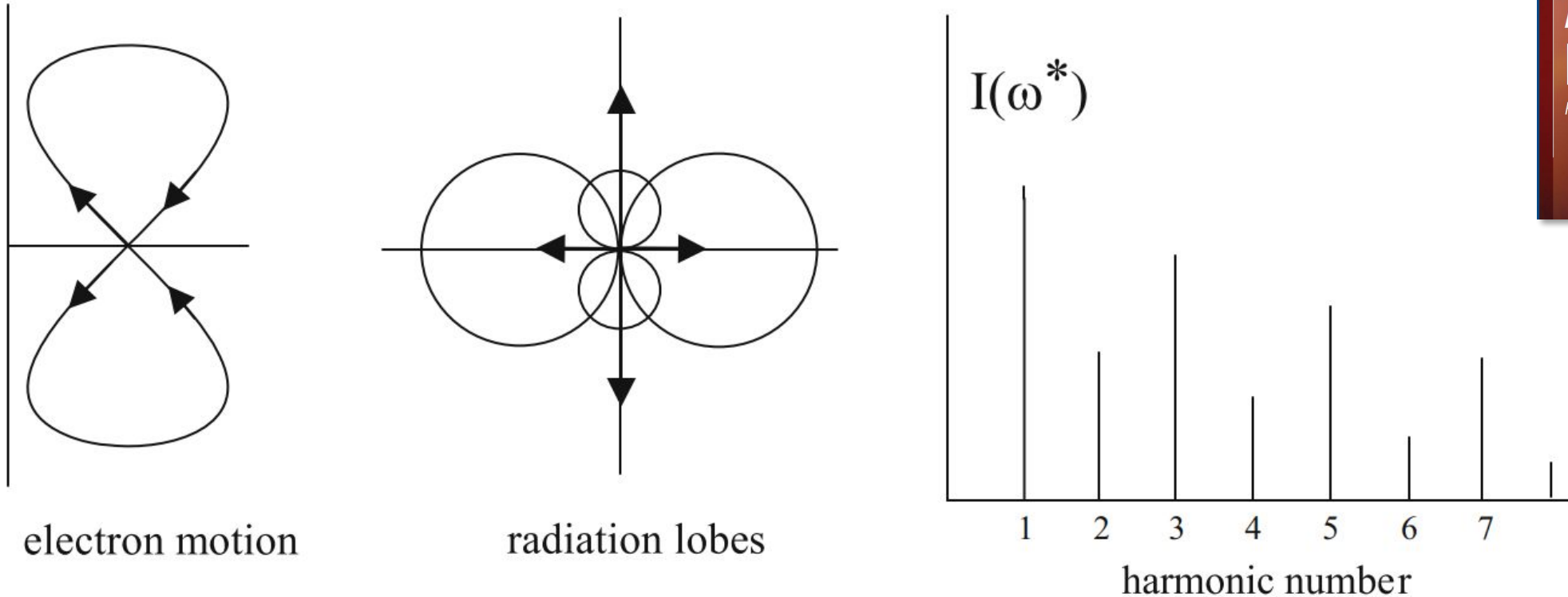
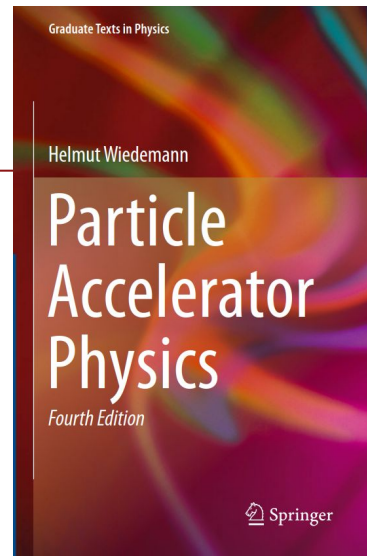


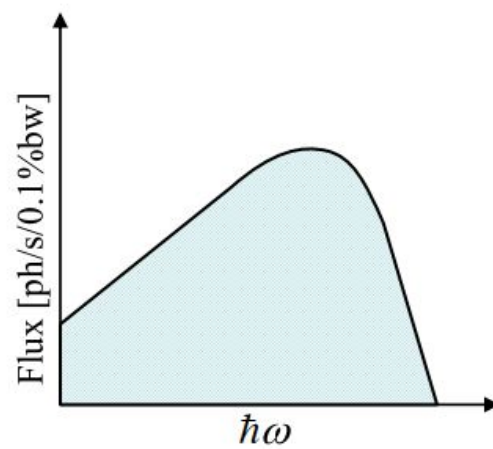
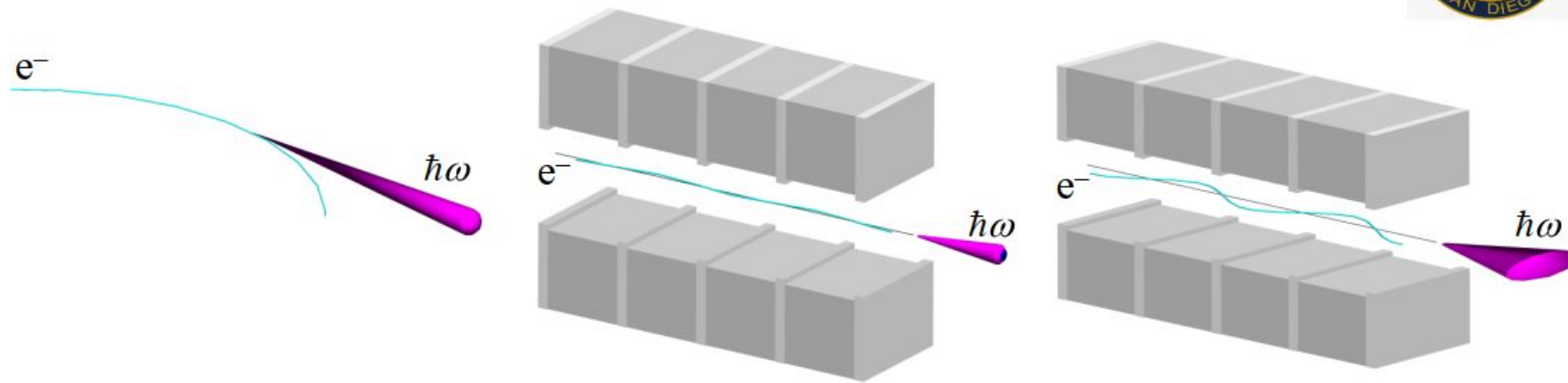
Fig. 24.8 Beam dynamics (*left*) and radiation lobes (*middle*) in the particle rest system together with the harmonics spectrum (*right*) for a stronger undulator ($K \gtrsim 1$)



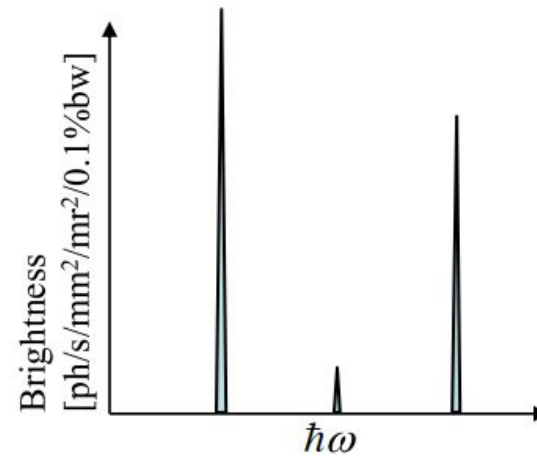
Bend

Undulator

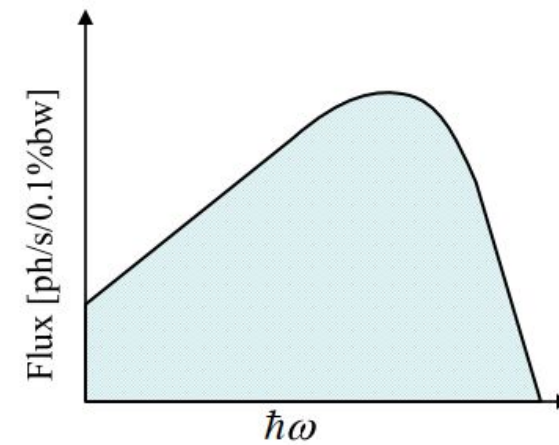
Wiggler



white source



partially coherent source



powerful white source



Damped synchrotron motion

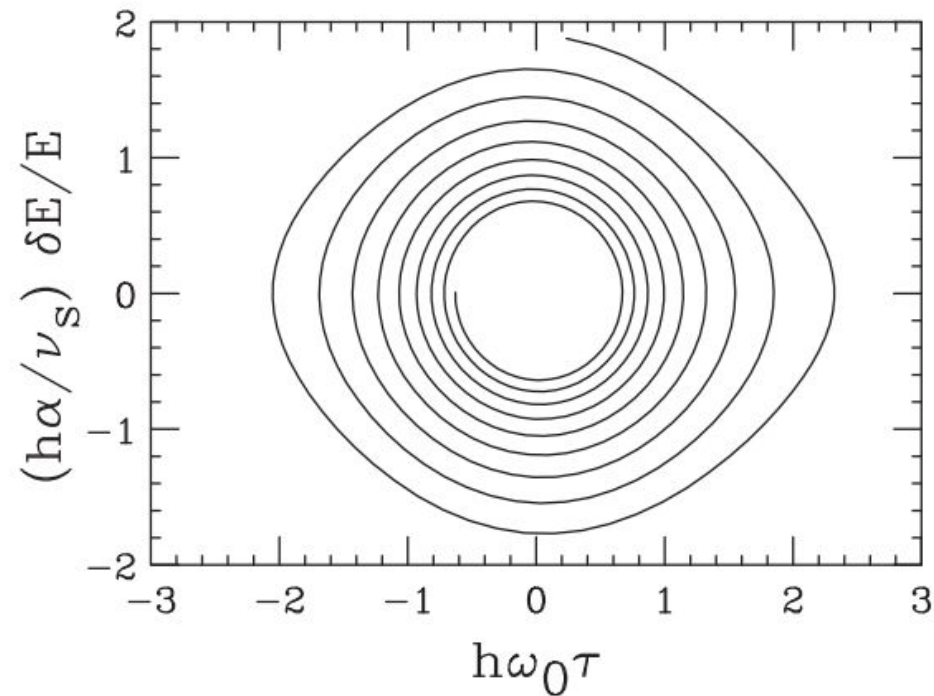
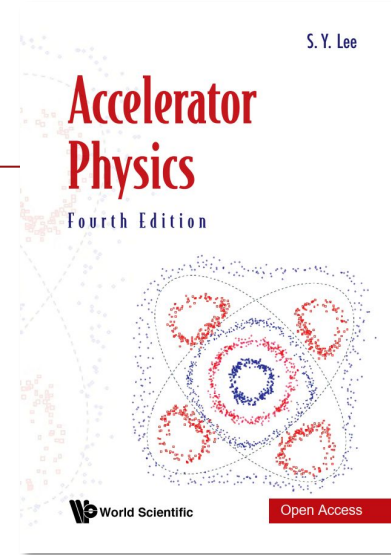
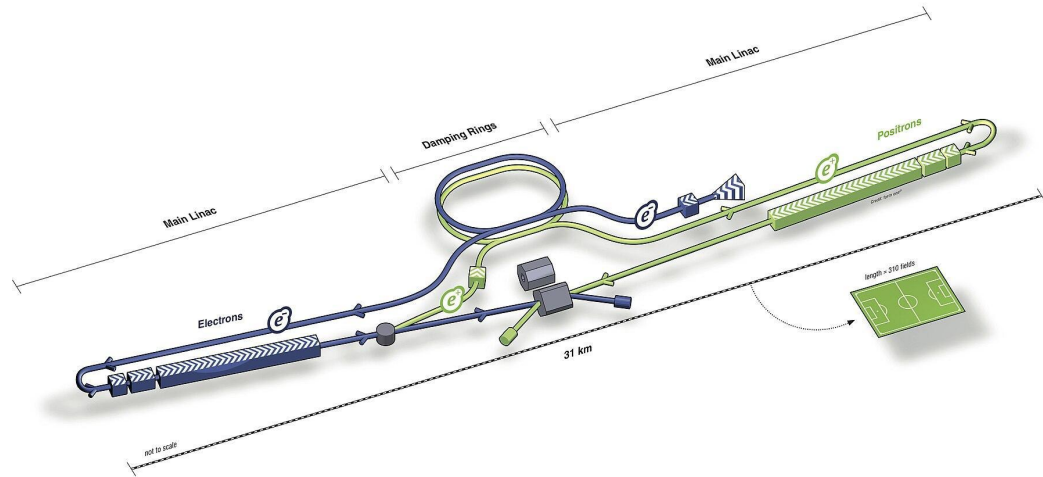


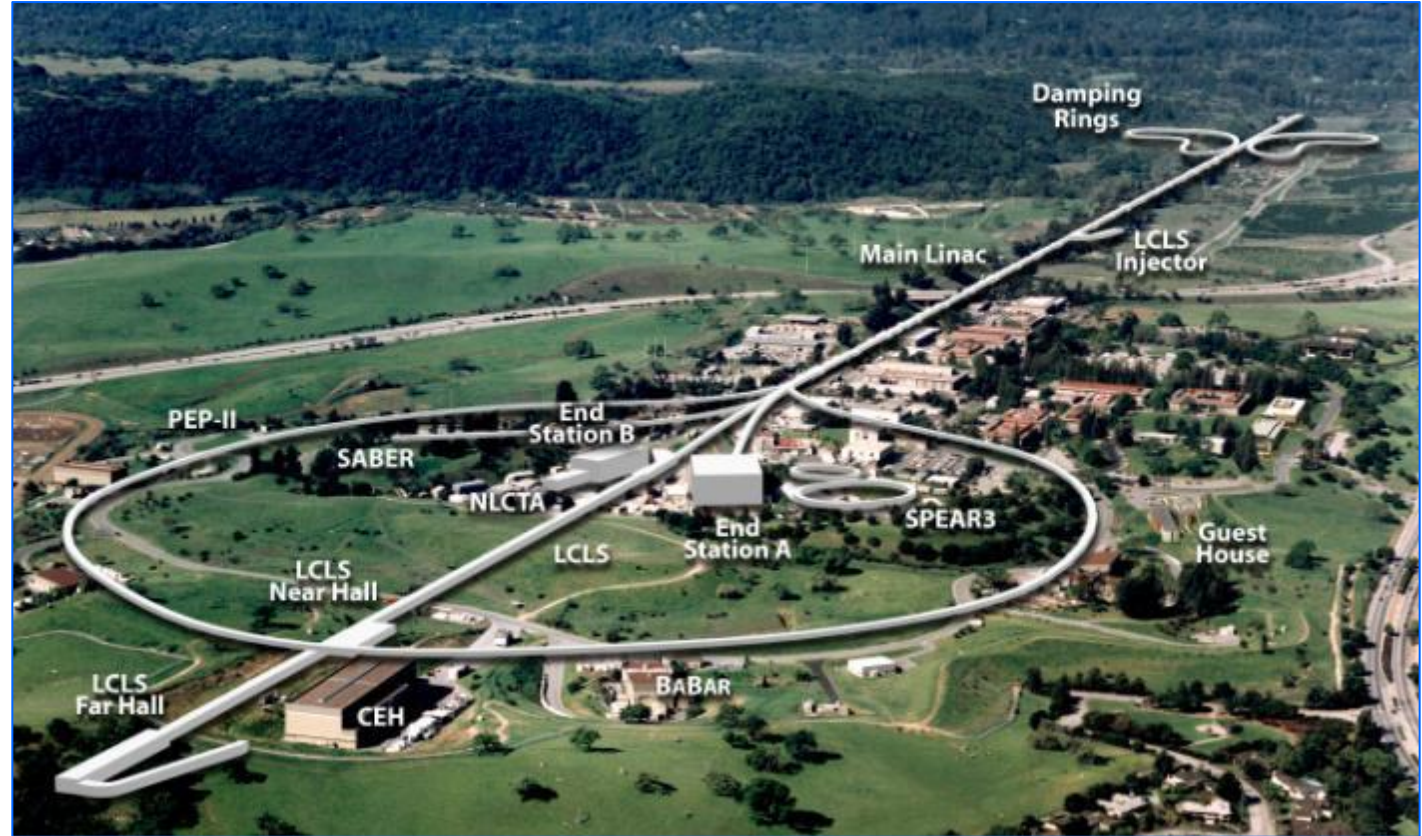
Figure 4.5: A schematic drawing of damped synchrotron motion. Particle motion is damped toward the center of the bucket.



Damping rings



ILC design



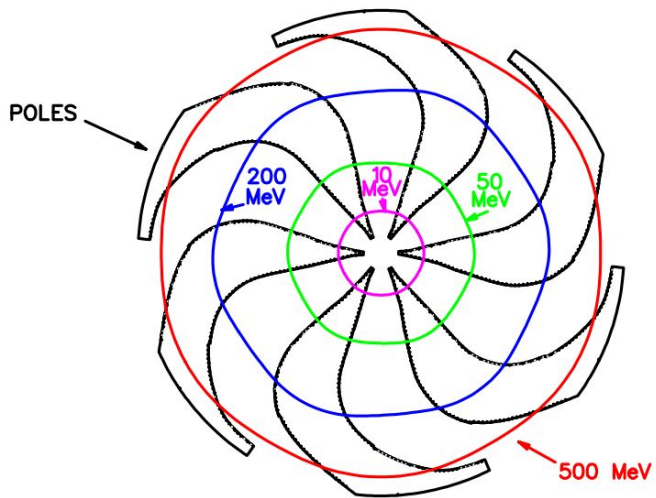
SLAC layout (mid-2000's ?)



TRIUMF

- 520 MeV H-
sector-bend
cyclotron

TRIUMF'S POLES AND EQUILIBRIUM ORBITS



Rick Baartman, TRIUMF

